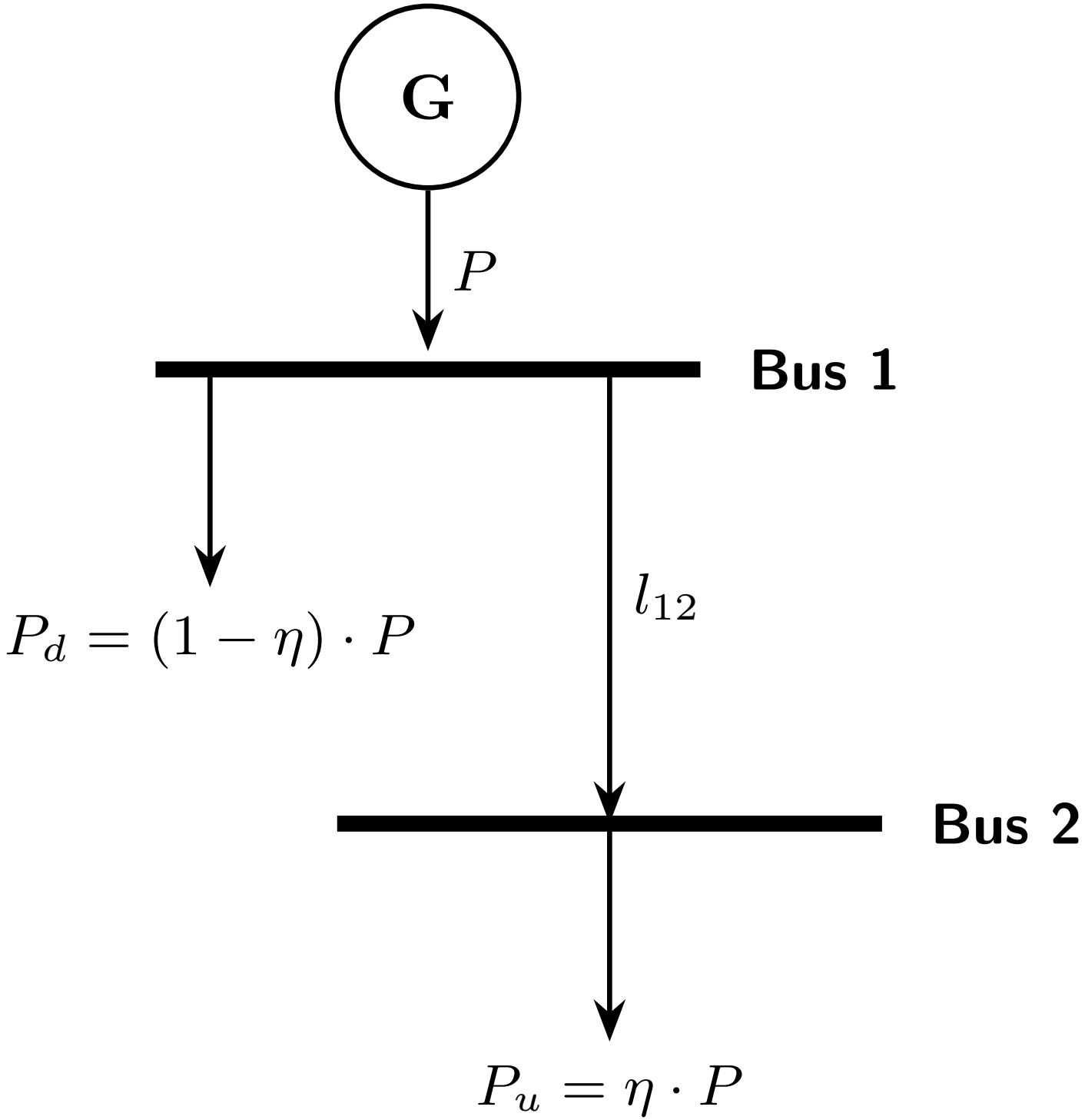

G
P
Bus 1
$P_d = (1 - \eta) \cdot P$
l_{12}
Bus 2
$P_u = \eta \cdot P$